

Mini Plants vs Zombies Game: Markdown Sheet

Decomposition

- **Classes:** Creating a "*Plant*" and "*Zombie*" classes with their own attributes and methods.
- **Objects:** Instantiate two "*Plant*" objects with their own damage values and one "*Zombie*" object.
- **Mechanics:** Calculating the damage, updating health values as they take damage.
- **Loop:** Make a loop until the "*Zombie*" dies or both Plants are eaten.
- **Output:** Display the updated stats after a turn and the final result.

Pattern Recognition

- **Turn-based:** Zombies and Plants take turns on who attacks.
- **Game end check:** At the start or end of the game, if the zombies health < 0 (plants win) if the plants health < 0 (zombies win)
- **Game Selection:** Check the Plant health so that the Zombie chooses to attack the one with lower health.
- **Check distance:** Check whether the distance is < 0 to decide if Zombie "*move()*" or "*attack(plant)*"

Abstraction

- Plants: name, damage, health
- Zombies: name, damage, health, distance

Algorithm

1. Initialize two Plant objects and one Zombie object with your starting stats.
2. Start the game loop
3. For each plant
 - i. Plants attack zombie using "*attack(zombie)*"
 - ii. If Zombie's health < 0 print "*Plants Win*" and end the game
4. For Zombie:
 - i. Identify the target plant
 - ii. If Zombie distance > 0 call method "*move()*", Otherwise if distance = 0 call "*attack(plant)*"
5. Check if the Plants health < 0 print "*Zombies Win!*", otherwise print Plants Win!
6. Repeat steps 3-5 until the Plants health reaches 0 or the Zombies health reaches 0.